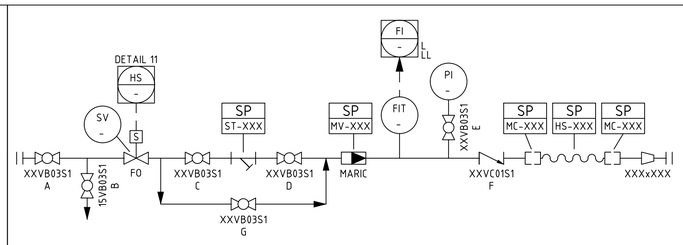
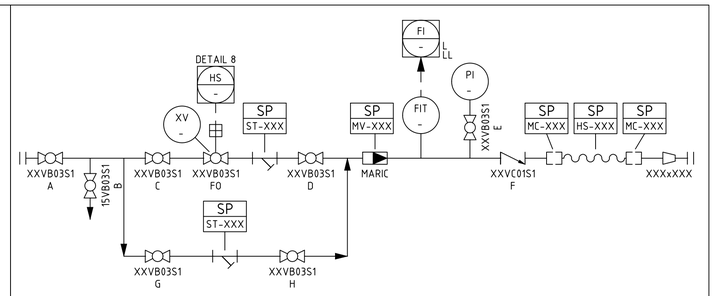


GLAND WATER DETAIL 1
SOLENOID VALVE AND FLOW SWITCH
(REFER TABLE FOR SIZING AND INFORMATION)



GLAND WATER DETAIL 2
SOLENOID VALVE AND FLOW TRANSMITTER (DN25 & UNDER)
(REFER TABLE FOR SIZING AND INFORMATION)



GLAND WATER DETAIL 3
BALL VALVE AND FLOW TRANSMITTER (DN32 & ABOVE)
(REFER TABLE FOR SIZING AND INFORMATION)

NOTES:

1. REFER TO 2000-F24-PID-PR-10003 AND 2000-F24-PID-PR-10014 FOR THE GREENFIELDS LOW PRESSURE GLAND SEAL WATER CIRCUIT.
2. REFER TO 2000-F24-PID-PR-10004 AND 2000-F24-PID-PR-10014 FOR THE GREENFIELDS HIGH PRESSURE GLAND SEAL WATER CIRCUIT.
3. PUMPS REQUIRING GLAND WATER THAT ARE WITHIN BROWNFIELDS SCOPE ARE TO TIE IN TO THE EXISTING GLAND WATER CIRCUIT. REFER TO 2000-F75-PID-PR-10007 FOR THE BROWNFIELDS GLAND WATER CIRCUIT.
4. VALVE SIZE MATCHES LINE SIZE UNLESS OTHERWISE SPECIFIED IN THE RELEVANT GLAND WATER DETAIL.
5. LP REFERS TO THE GREENFIELDS LOW PRESSURE GLAND WATER CIRCUIT, HP REFERS TO THE GREENFIELDS HIGH PRESSURE GLAND WATER CIRCUIT AND BF REFERS TO THE BROWNFIELDS GLAND WATER CIRCUIT.
6. GLAND WATER CONNECTION TO PUMPS F24-PW-401/402 IS A DN25 FLANGED CONNECTION. CONNECTING FLANGE DRILLED TO SUIT DN25 TBE PIPE NIPPLE FOR MINSUP CONNECTION.
7. REDUCING BUSH REQUIRED BETWEEN HOSE AND PUMP CONNECTION.

EQUIPMENT DETAILS						MANUAL VALVE TAG NUMBERS								INSTRUMENTATION			SPECIAL PIPING ITEMS				
EQUIPMENT NUMBER	PUMP DESCRIPTION	P&ID NUMBER	LINE SIZE NOTE 4	DETAIL	HP/LP/BF NOTE 5	A	B	C	D	E	F	G	H	XV/SV/HS	FSL/FAL	PI	SP-MV	SP-ST	SP-MC	SP-HS	REDUCER
F14-PP-421	INTENSIVE CYANIDE LEACH REACTOR DISCHARGE PUMP	2000-F14-PID-PR-10003	25	1	LP	F14-13151	F14-13152	F14-13153	F14-13154	F14-13156	F14-13157	F14-13155	-	14118	14119	14120	-	011	001	015	25x15
F16-PP-515	FLOTATION CELL DRAIN PUMP	2000-F16-PID-PR-10003	25	1	LP	F16-13158	F16-13159	F16-13160	F16-13161	F16-13163	F16-13164	F16-13162	-	16357	16358	16359	004	011	001	015	25x15
F16-PP-638	SAMPLE RETURN PUMP 1	2000-F16-PID-PR-10005	25	1	LP	F16-13165	F16-13166	F16-13167	F16-13168	F16-13170	F16-13171	F16-13169	-	16155	16156	16157	001	011	001	015	25x8
F16-PP-639	SAMPLE RETURN PUMP 2	2000-F16-PID-PR-10005	25	1	LP	F16-13172	F16-13173	F16-13174	F16-13175	F16-13177	F16-13178	F16-13176	-	16165	16166	16167	001	011	001	015	25x8
F18-PP-407	FINAL CONCENTRATE PUMP 1	2000-F18-PID-PR-10001	25	1	LP	F18-13179	F18-13180	F18-13181	F18-13182	F18-13184	F18-13185	F18-13183	-	18500	18501	18502	004	011	001	015	25x15
F18-PP-408	FINAL CONCENTRATE PUMP 2	2000-F18-PID-PR-10001	25	1	LP	F18-13186	F18-13187	F18-13188	F18-13189	F18-13191	F18-13192	F18-13190	-	18504	18505	18506	004	011	001	015	25x15
F18-PP-501	CLEANER FLOTATION CELL RECIRCULATION PUMP 1	2000-F18-PID-PR-10001	25	1	LP	F18-13193	F18-13194	F18-13195	F18-13196	F18-13198	F18-13199	F18-13197	-	18508	18509	18510	002	011	001	015	N/A
F18-PP-502	CLEANER FLOTATION CELL RECIRCULATION PUMP 2	2000-F18-PID-PR-10001	25	1	LP	F18-13200	F18-13201	F18-13202	F18-13203	F18-13205	F18-13206	F18-13204	-	18512	18513	18514	002	011	001	015	N/A
F18-PP-511	CLEANER-SCAVENGER FLOTATION CELL RECIRCULATION PUMP 1	2000-F18-PID-PR-10003	25	1	LP	F18-13207	F18-13208	F18-13209	F18-13210	F18-13212	F18-13213	F18-13211	-	18524	18525	18526	002	011	001	015	N/A
F18-PP-512	CLEANER-SCAVENGER FLOTATION CELL RECIRCULATION PUMP 2	2000-F18-PID-PR-10003	25	1	LP	F18-13214	F18-13215	F18-13216	F18-13217	F18-13219	F18-13220	F18-13218	-	18528	18529	18530	002	011	001	015	N/A
F18-PP-513	CLEANER-SCAVENGER FLOTATION CELL TAILS PUMP 1	2000-F18-PID-PR-10003	25	1	LP	F18-13221	F18-13222	F18-13223	F18-13224	F18-13226	F18-13227	F18-13225	-	18532	18533	18534	002	011	001	015	N/A
F18-PP-514	CLEANER-SCAVENGER FLOTATION CELL TAILS PUMP 2	2000-F18-PID-PR-10003	25	1	LP	F18-13228	F18-13229	F18-13230	F18-13231	F18-13233	F18-13234	F18-13232	-	18536	18537	18538	002	011	001	015	N/A
F24-PW-401	MILL WATER PUMP 1	2000-F24-PID-PR-10001	25	1	LP	F24-13235	F24-13236	F24-13237	F24-13238	F24-13240	F24-13241	F24-13239	-	24512	24510	24511	002	011	001	015	NOTE 6
F24-PW-402	MILL WATER PUMP 2	2000-F24-PID-PR-10001	25	1	LP	F24-13242	F24-13243	F24-13244	F24-13245	F24-13247	F24-13248	F24-13246	-	24515	24513	24514	002	011	001	015	NOTE 6
F13-PP-401	CYCLONE FEED PUMP	2000-F13-PID-PR-10002	40	3	HP	F13-10900	F13-10901	F13-10902	F13-10903	F13-10906	F13-10907	F13-10904	F13-10905	13460	FIT13461	13462	003	012	011	017	40x32
F13-PP-403	TRASH DEWATERING SCREEN UNDERFLOW PUMP	2000-F13-PID-PR-10007	25	1	HP	F13-10908	F13-10909	F13-10910	F13-10911	F13-10913	F13-10914	F13-10912	-	13464	13465	13466	005	011	001	015	25x8
F17-PP-455	FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 1	2000-F17-PID-PR-10002	25	2	HP	F17-10915	F17-10916	F17-10917	F17-10918	F17-10920	F17-10921	F17-10919	-	17104	FIT17105	17106	002	011	001	015	N/A
F17-PP-456	FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 2	2000-F17-PID-PR-10002	25	2	HP	F17-10922	F17-10923	F17-10924	F17-10925	F17-10927	F17-10928	F17-10926	-	17114	FIT17115	17116	002	011	001	015	N/A
F23-PP-466	FINAL TAILS PUMP 1	2000-F23-PID-PR-10002	25	2	HP	F23-10929	F23-10930	F23-10931	F23-10932	F23-10934	F23-10935	F23-10933	-	23480	FIT23481	23482	002	011	001	015	N/A
F23-PP-467	FINAL TAILS PUMP 2	2000-F23-PID-PR-10002	25	2	HP	F23-10936	F23-10937	F23-10938	F23-10939	F23-10941	F23-10942	F23-10940	-	23484	FIT23485	23486	002	011	001	015	N/A
F23-PP-468	FINAL TAILS PUMP 3	2000-F23-PID-PR-10002	25	2	HP	F23-10943	F23-10944	F23-10945	F23-10946	F23-10948	F23-10949	F23-10947	-	23488	FIT23489	23490	002	011	001	015	N/A
F23-PP-469	FINAL TAILS PUMP 4	2000-F23-PID-PR-10002	25	2	HP	F23-10950	F23-10951	F23-10952	F23-10953	F23-10955	F23-10956	F23-10954	-	23492	FIT23493	23494	002	011	001	015	N/A
F24-PW-417	HIGH PRESSURE PROCESS WATER PUMP 1	2000-F24-PID-PR-10001	25	1	HP	F24-10957	F24-10958	F24-10959	F24-10960	F24-10962	F24-10963	F24-10961	-	24518	24516	24517	006	011	001	015	N/A
F24-PW-418	HIGH PRESSURE PROCESS WATER PUMP 2	2000-F24-PID-PR-10001	25	1	HP	F24-10964	F24-10965	F24-10966	F24-10967	F24-10969	F24-10970	F24-10968	-	24521	24519	24520	006	011	001	015	32x25
F28-PP-409	UFG PRODUCT PUMP 1	2000-F28-PID-PR-10005	25	1	HP	F28-10971	F28-10972	F28-10973	F28-10974	F28-10976	F28-10977	F28-10975	-	28400	28401	28402	004	011	001	015	25x15
F28-PP-410	UFG PRODUCT PUMP 2	2000-F28-PID-PR-10005	25	1	HP	F28-10978	F28-10979	F28-10980	F28-10981	F28-10983	F28-10984	F28-10982	-	28404	28405	28406	004	011	001	015	25x15
F28-PP-453	UFG MILL 2 DISCHARGE PUMP	2000-F28-PID-PR-10002	25	1	HP	F28-11900	F28-11901	F28-11902	F28-11903	F28-11905	F28-11906	F28-11904	-	28412	28413	28414	004	011	001	015	25x15
F28-PP-451	UFG MILL 2 FEED PUMP	2000-F28-PID-PR-10001	25	1	HP	F28-11914	F28-11915	F28-11916	F28-11917	F28-11919	F28-11920	F28-11918	-	28408	28409	28410	005	011	001	015	25x8
F28-PW-421	UFG MILL 2 ISACHARGER PUMP	2000-F28-PID-PR-10001	25	1	HP	F28-11928	F28-11929	F28-11930	F28-11931	F28-11933	F28-11934	F28-11932	-	28416	28417	28418	002	011	001	015	25x10
F30-PP-417	CONCENTRATE THICKENER U/F PUMP 1	2000-F30-PID-PR-10002	25	1	BF	F30-13249	F30-13250	F30-13251	F30-13252	F30-13254	F30-13255	F30-13253	-	30447	30448	30449	001	011	001	015	25x8
F30-PP-418	CONCENTRATE THICKENER U/F PUMP 2	2000-F30-PID-PR-10002	25	1	BF	F30-13256	F30-13257	F30-13258	F30-13259	F30-13261	F30-13262	F30-13260	-	30452	30453	30454	001	011	001	015	25x8
F35-PP-448	UFG CIRCUIT 2 FEED PUMP	2000-F35-PID-PR-10002	25	1	BF	F35-13263	F35-13264	F35-13265	F35-13266	F35-13268	F35-13269	F35-13267	-	35437	35438	35439	001	011	001	015	25x8

DRG No.	REFERENCES
2000-F00-PID-PR-10007	LEGEND SHEET 7 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10006	LEGEND SHEET 6 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10005	LEGEND SHEET 5 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10004	LEGEND SHEET 4 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10003	LEGEND SHEET 3 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10002	LEGEND SHEET 2 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10001	LEGEND SHEET 1 - PIPING AND INSTRUMENTATION DIAGRAM

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
0	25MAR25	APPROVED FOR CONSTRUCTION	[redacted]				

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: AIR & WATER SERVICES
GLAND WATER - SHEET 4
PIPING AND INSTRUMENTATION DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NTS	2000-F24-PID-PR-10034	0

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